Activity # 15: SOC Log Sanitization & Attack Detection

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Cyber Handle: MaskedLogic

1. Why are messy logs dangerous in cybersecurity analysis? Messy logs are dangerous because they obscure real threats. This cause slows down an analyst's response.

2. Which types of logs did you remove and why? I removed routine operational logs such as [INFO], User login OK, and random string noises Random text here, Noise noise noise. These were removed because they represent normal network behavior.

3. Why do analysts focus on "Failed password" logs first? Analysts prioritize "Failed password" logs because multiple failed attempts are the most basic and direct brute-force attacks.

4. Why is log normalization important before analysis? Log normalization standardizes the format of logs (replacing various warnings with a standard [ALERT] tag).

5. What pattern is used to identify an IP address? A Regular Expression (Regex) pattern is used. Specifically, the pattern [0-9]+\.[0-9]+\.[0-9]+\.[0-9]+ identifies an IPv4 address.

6. Which IP appears most frequently? The IP address **203.0.113.50** appears most frequently, with a total of 4 failed password attempts.

7. Which IP is performing a brute force attack? The IPs **10.0.0.5** (3 attempts) and **203.0.113.50** (4 attempts) are performing brute-force attacks.

8. Why is masking sensitive data important in reports? Data masking like replacing IPs with XXX.XXX.XXX.XXX protects Personally Identifiable Information and ensures compliance with data privacy regulations.

9. How does combining multiple replacements improve efficiency? Combining replacements changing Failed to [ALERT] and Accepted to [SUCCESS] in one command improves efficiency by processing the log file in a single pass.

10. What advantage does real-time monitoring give a SOC analyst? Real-time monitoring allows an analyst to detect and observe threats the exact moment they occur. This immediate visibility provides the crucial time needed to block attacker IPs, isolate affected systems, and mitigate the attack before any data compromise.

```
ERROR: Failed password from XXX.XXX.XXX.XXX
User login OK
```

```
root@kali: ~

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C:\Users\User>ssh kali@192.168.126.129
kali@192.168.126.129's password:
Linux kali 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon May 25 20:36:54 2026 from 192.168.126.1
┌──(kali㉿kali)-[~]
└─$ sudo -i
[sudo] password for kali:
┌──(root㉿kali)-[~]
└─# echo "Failed password from 45.33.32.1" >> messy.log

┌──(root㉿kali)-[~]
└─#
```

```
┌──(root㉿kali)-[~]
└─# tail -f messy.log
ERROR: Failed password from 10.0.0.5
User login OK
Failed password from 10.0.0.5
[ALERT] Failed password from 10.0.0.5
Noise noise noise
Accepted login from 203.0.113.50
Failed password from 203.0.113.50
Failed password from 203.0.113.50
Failed password from 203.0.113.50
#FLAG{log_cleaning_started}
Failed password from 45.33.32.1
```